AWS Networking Concepts Documentation

1. Introduction

Modern cloud architectures require scalable, secure, and well-structured networking. AWS provides various networking components such as **subnets, VPC peering, and transit gateways** to build such architectures.

This document explains key networking concepts used in multi-VPC environments, including:

- Public vs Private Subnets
- VPC Peering vs Transit Gateway
- Benefits of Multi-VPC Architecture

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2. Public Subnet vs Private Subnet

2.1 Public Subnet

A **public subnet** is a subnet that has direct access to the internet through an **Internet Gateway (IGW)**.

Characteristics

- Connected to Internet Gateway
- Instances can have public IP addresses
- Direct internet access

Use Cases

- Web servers
- Bastion hosts
- Load balancers
- NAT Gateways

2.2 Private Subnet

A **private subnet** does not have direct access to the internet.

Characteristics

- No direct connection to Internet Gateway
- Instances usually have private IPs only
- Internet access (if needed) via NAT Gateway

Use Cases

- Application servers
- Databases
- Backend services

2.3 Key Differences

Feature	Public Subnet	Private Subnet
Internet Access	Direct	Indirect (via NAT)
IP Type	Public + Private	Private only
Security	Less secure	More secure
Use Case	Frontend services	Backend systems

Simple Explanation

Public Subnet = Internet-facing
Private Subnet = Internal-only

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3. VPC Peering vs Transit Gateway

3.1 VPC Peering

VPC Peering is a **direct connection between two VPCs**.

Characteristics

- Point-to-point connection
- No transitive routing

- Simple and low-cost

Use Cases

- Connecting two VPCs
- Small-scale architectures

Limitations

- Does not scale well
- Complex when many VPCs are involved

3.2 Transit Gateway

Transit Gateway acts as a **central hub to connect multiple VPCs and networks**.

Characteristics

- Hub-and-spoke architecture
- Supports transitive routing
- Scalable and centralized

Use Cases

- Large multi-VPC environments
- Multi-region architectures
- Enterprise networking

3.3 Key Differences

Feature	VPC Peering	Transit Gateway
Type	Point-to-point	Hub-based
Scalability	Limited	Highly scalable
Routing	No transitive routing	Supports transitive routing
Complexity	Simple	Centralized management
Cost	Lower	Higher

Simple Explanation

VPC Peering = Direct connection (like a cable between two systems)
Transit Gateway = Central router connecting many networks

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4. Benefits of Multi-VPC Architecture

4.1 Improved Security

- Isolation between environments
- Separate VPCs for production, staging, and development

Reduces risk of security breaches

4.2 Better Scalability

- Easy to add new VPCs as system grows
- Supports large distributed systems

4.3 Fault Isolation

- Issues in one VPC do not affect others

Improves system reliability

4.4 Access Control

- Fine-grained control over communication between VPCs
- Use route tables and security groups

4.5 Multi-Region Deployment

- Deploy services across different regions
- Improves availability and performance

4.6 Flexible Network Design

- Custom CIDR ranges

- Different subnet strategies per VPC

4.7 Organizational Separation

- Different teams can manage different VPCs
- Useful for large organizations

4.8 Performance Optimization

- Traffic routing can be optimized
- Reduced congestion

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5. Conclusion

- Public and private subnets help separate internet-facing and internal resources
- VPC Peering is suitable for small setups, while Transit Gateway is ideal for large architectures
- Multi-VPC architecture provides better security, scalability, and flexibility

These concepts are essential for designing modern cloud network architectures.

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6. Video Explanation